In silico analysis of hypothetical protein (YP_005229762.1) from *Klebsiella pneumoniae*

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Dr. Jannatul Ferdoush

Introduction

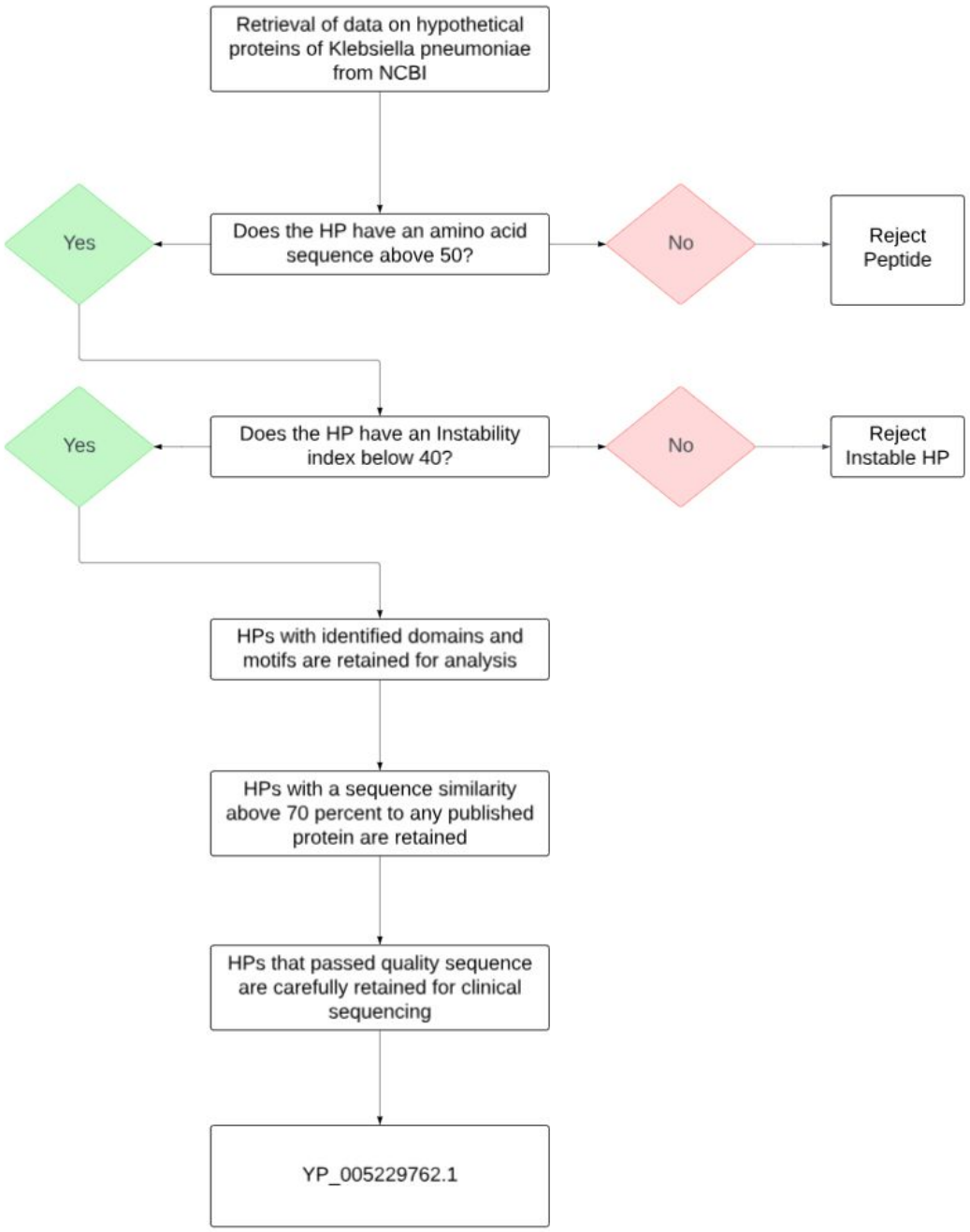
- Overview of Klebsiella pneumoniae
- Importance of studying hypothetical proteins (HP)
- Antibiotic resistance issue

Background

- Next-generation sequencing (NGS) and identification of HPs
- Importance of in silico characterization
- Potential applications of HPs in drug design and discovery.

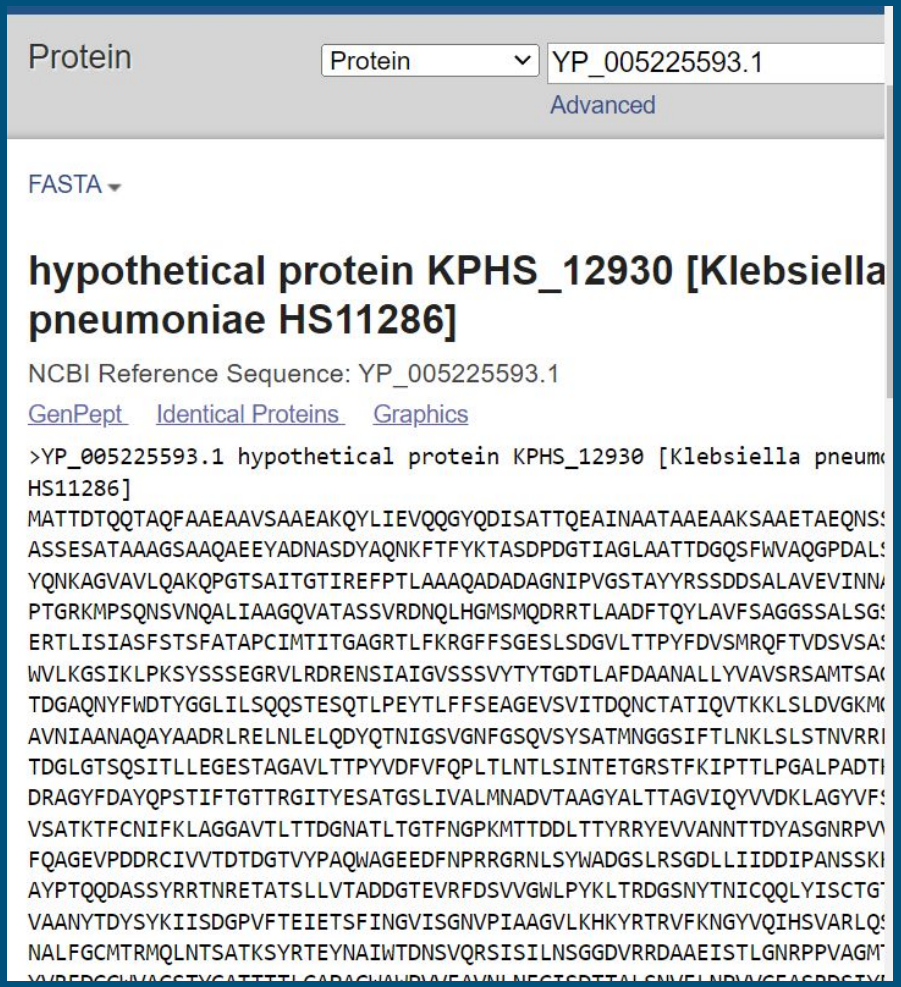
Objective

Investigate and characterize a hypothetical protein that would be the best fit for drug screening and discovery to help aid Klebsiella pneumoniae.



Data Collection

- Source: National Center for Biotechnology Information (NCBI) database
- Initial selection of HPs
- Filtering out peptides (sequences shorter than 50 residues)



807 HPs

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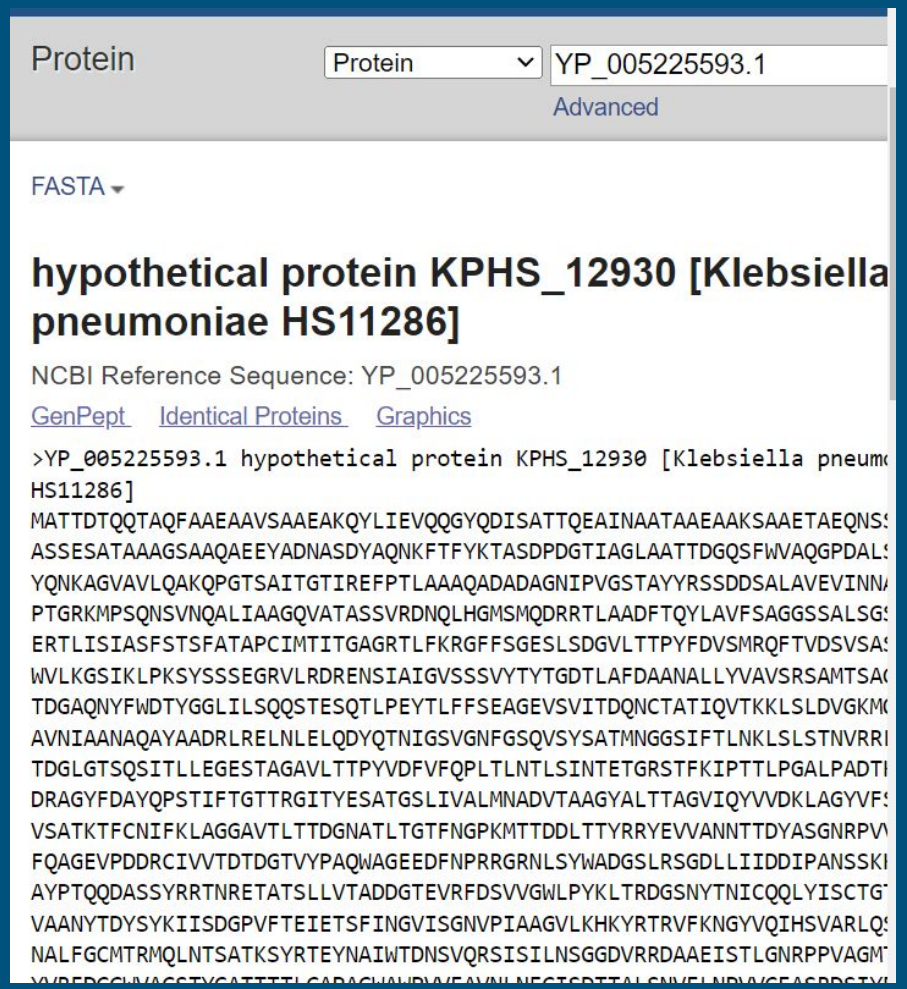

807 HPs → **557 HPs**

ProtParam Analysis

Expasy
Swiss Bioinformatics Resource Portal

- Use of ProtParam tool to analyze physicochemical properties.
- Parameters: Molecular weight, theoretical pI, amino acid composition, etc.
- Selection based on stability (instability index). A value under 40 is considered stable.

Descriptions	Value
Number of amino acids	139
Molecular Weight	15812.05 KDa
Theoretical pI	4.56
Total number of negatively charged residues	25
Total number of positively charged residues	15
Ext. coefficient	20970 M^{-1} cm^{-1}
Instability index	31.4
Aliphatic index	99.64
Grand average of hydropathicity (GRAVY)	-0.267

$$Instability\ Index = \frac{10}{L}\sum_{i=1}^{L-1} DIWV(x_i, x_{i+1})$$

ProtParam

ProtParam [Documentation / Reference] is a tool which allows the computation of various physical and chemical parameters for a given protein stored in UniProtKB or for a user entered protein sequence. The computed parameters include the molecular weight, theoretical pI, amino acid composition, atomic composition, extinction coefficient, estimated half-life, instability index, aliphatic index and grand average of hydropathicity (GRAVY) (Disclaimer).

Enter a protein sequence

Please enter one UniProtKB AC/ID (e.g. *P05130* or *KPC1_DROME*).
Alternatively, enter one protein sequence in single letter code (e.g. *ABCDEFGHIKLMNOPQRSTUVWXY*).

Examples

MATTDTQQTAQFAAEAAVSAAEAKQYLIEVQQGYQDISATTQEAINAATAAEA
AKSAAETAEQNSSTSAA
ASSESATAAAGSAAQAEEYADNASDYAQNKFTFYKTASDPDGTIAGLAATTD
GQSFWVAQGPDALSAAWQ
YQNKAGVAVLQAKQPGTSAITGTIREFPTLAAAQADADAGNIPVGSTAYYRSS
DDSALAVEVINNAGTLQ
PTGRKMPSQNSVNQALIAAGQVATASSVRDNQLHGMSMQDRRTLAADFTQY
LAVFSAGGSSALSGSGTYA

RESET | Compute parameters

557 HPs

Instability index:
The instability index (II) is computed to be 31.90
This classifies the protein as stable.

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MATTDTQQTAQFAAEAAVSAAEAKQYLIEVQQGYQDISATTQEAINAATAAEA
AKSAAETAEQNSSTSAA
ASSESATAAAGSAAQAEEYADNASDYAQNKFTFYKTASDPDGTIAGLAATTD
GQSFWVAQGPDALSAAWQ
YQNKAGVAVLQAKQPGTSAITGTIREFPTLAAAQADADAGNIPVGSTAYYRSS
DDSALAVEVINNAGTLQ
PTGRKMPSQNSVNQALIAAGQVATASSVRDNQLHGMSMQDRRTLAADFTQY
LAVFSAGGSSALSGSGTYA

RESET | Compute parameters

557 HPs → 274 HPs

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The instability index (II) is computed to be 31.90
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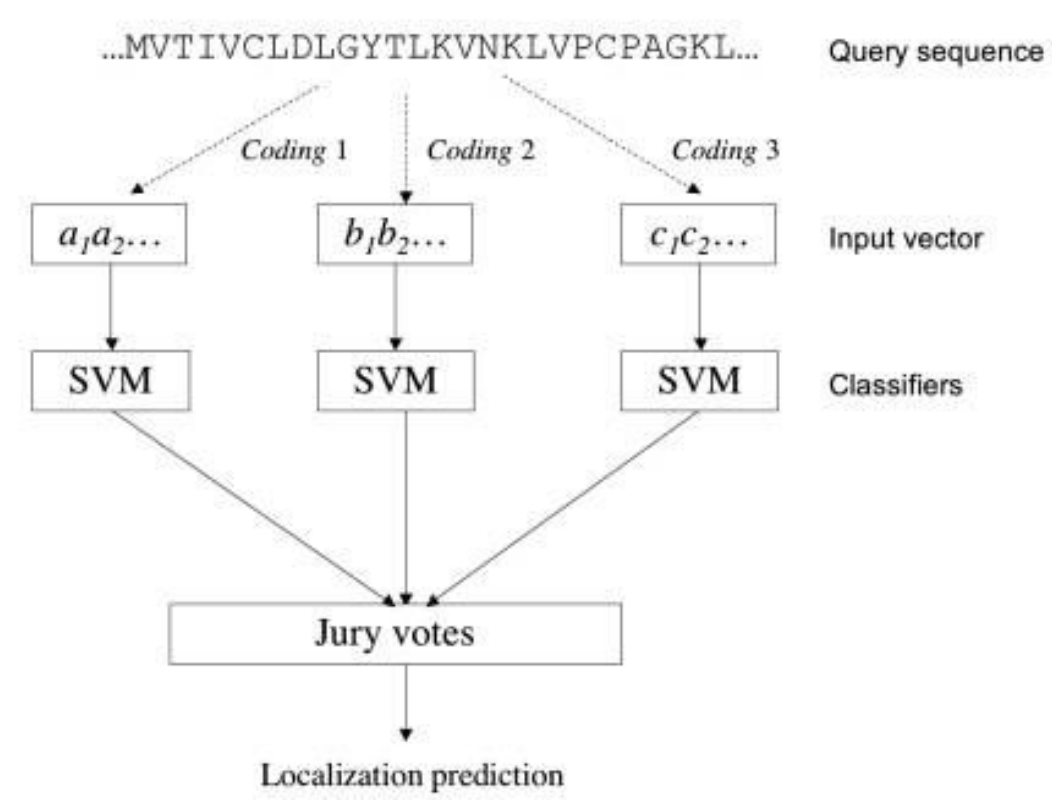
Subcellular Localization

- Tools: CELLO v.2.5 and PSortB v3.0.2
- Cross-referencing results for consistency
- Tools use support vector machines (SVMS) in predicting subcellular localization



SeqID: YP_005229762.1 hypothetical protein (plasmid) [Klebsiella pneumoniae subsp. pneumoniae HS11286]

Analysis Report:

SVM	LOCALIZATION	RELIABILITY
Amino Acid Comp.	Cytoplasmic	0.810
N-peptide Comp.	Cytoplasmic	0.895
Partitioned seq. Comp.	Cytoplasmic	0.903
Physico-chemical Comp.	Cytoplasmic	0.964
Neighboring seq. Comp.	Cytoplasmic	0.959

CELLO Prediction:

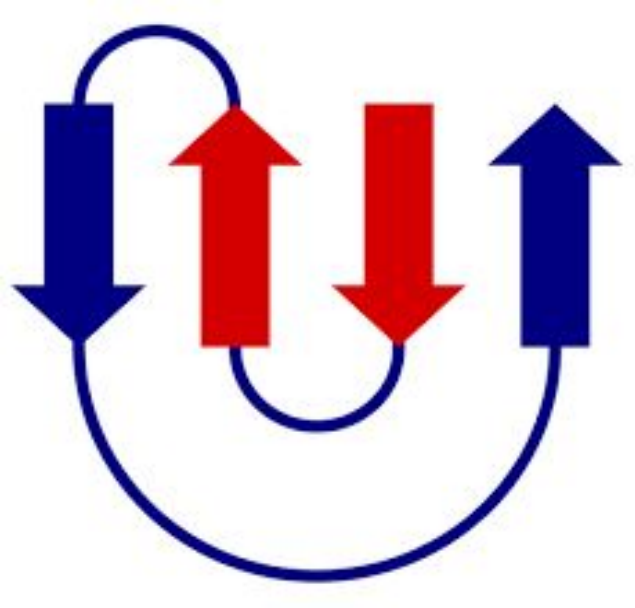
	Cytoplasmic	4.529 *
	InnerMembrane	0.227
	Periplasmic	0.162
	Extracellular	0.047
	OuterMembrane	0.035

PSORTb Results (Click here for an explanation of the output formats)

```
SeqID: YP_005229762.1 hypothetical protein (plasmid) [Klebsiella pneumoniae subsp. pneumoniae HS11286]
  Analysis Report:
    CMSVM-            Unknown                      [No details]
    CytoSVM-          Cytoplasmic                  [No details]
    ECSVM-            Unknown                      [No details]
    ModHMM-           Unknown                      [No internal helices found]
    Motif-            Unknown                      [No motifs found]
    OMPMotif-         Unknown                      [No motifs found]
    OMSVM-            Unknown                      [No details]
    PPSVM-            Unknown                      [No details]
    Profile-          Unknown                      [No matches to profiles found]
    SCL-BLAST-        Unknown                      [No matches against database]
    SCL-BLASTe-       Unknown                      [No matches against database]
    Signal-           Unknown                      [No signal peptide detected]
  Localization Scores:
    Cytoplasmic            8.96
    CytoplasmicMembrane    0.51
    Periplasmic            0.26
    OuterMembrane          0.01
    Extracellular          0.26
  Final Prediction:
    Cytoplasmic            8.96
--------------------------------------------------------------------------------
```

Domain and Motif Identification

- Use of NCBI Batch CD-Search for domain identification
- Use of Motif search tool for motif identification
- Filtering based on known domains and motifs



274 HPs

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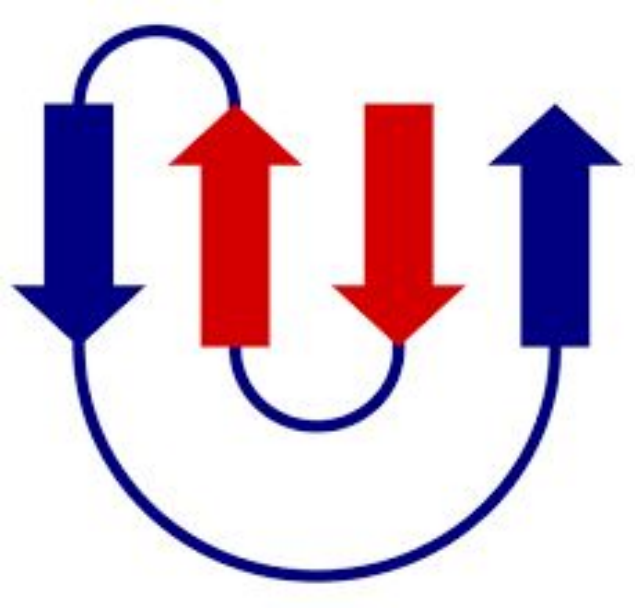

274 HPs → **71 HPs**

Homology Modeling

Select structural/domain databases

PDB_mmCIF70_8_Mar
Pfam-A_v37
UniProt-SwissProt-viral7...
NCBI_Conserved_Domai...

- 3D structure prediction of YP_005229762.1 with HHPred.
- Homology with published protein structures.
- Look for HPs with at least 70% sequence similarity to already published proteins.

Template alignment | CDD

1. PF19376.4 ; DUF5951 ; Family of unknown function (DUF5951)

Probability: 99.98%, E-value: 1.6e-33, Score: 163.42, Aligned cols: 50, Identities: 100%, Similarity: 1.376, Template Neff: 2.5

71 HPs

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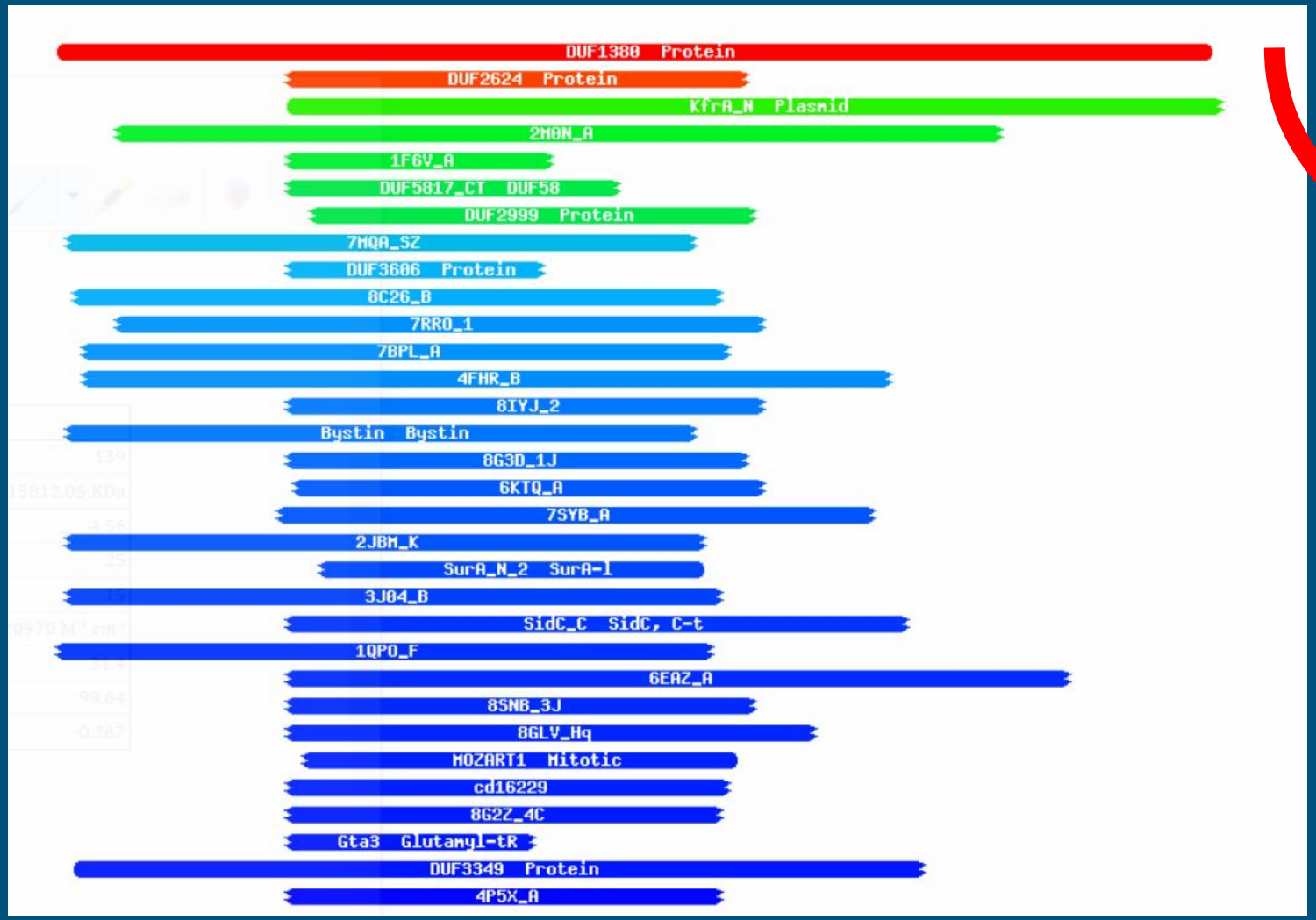
71 HPs → **7 HPs**



DUF1380 domain-containing protein

GenPept FASTA Graphics BLAST

Name: DUF1380 domain-containing protein
RefSeq Selected Product: WP_001568043.1, 139 amino acids
Assembly Accessions: 28349
Protein Accessions: 21097
CDS Regions: 30077
Total Rows: 30077

Model 01

Structure Assessment

Compare ☐

Download files ▾

Oligo-State ?
Monomer

GMQE ?
0.86

Template

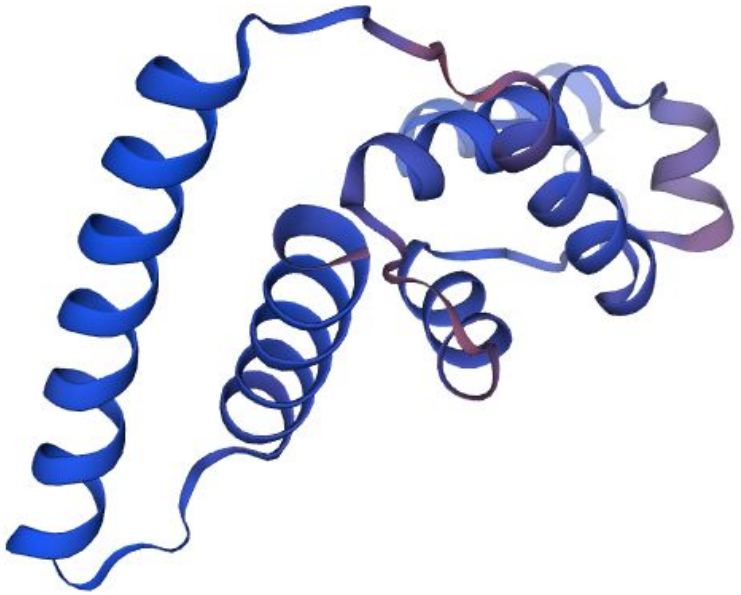
A0A0H3H1K3.1.A DUF1380 domain-containing protein

AlphaFold DB model of A0A0H3H1K3_KLEPH (gene: A0A0H3H1K3_KLEPH, organism: Klebsiella pneumoniae subsp pneumoniae (strain HS11286))

Seq Identity
100.00%

Coverage

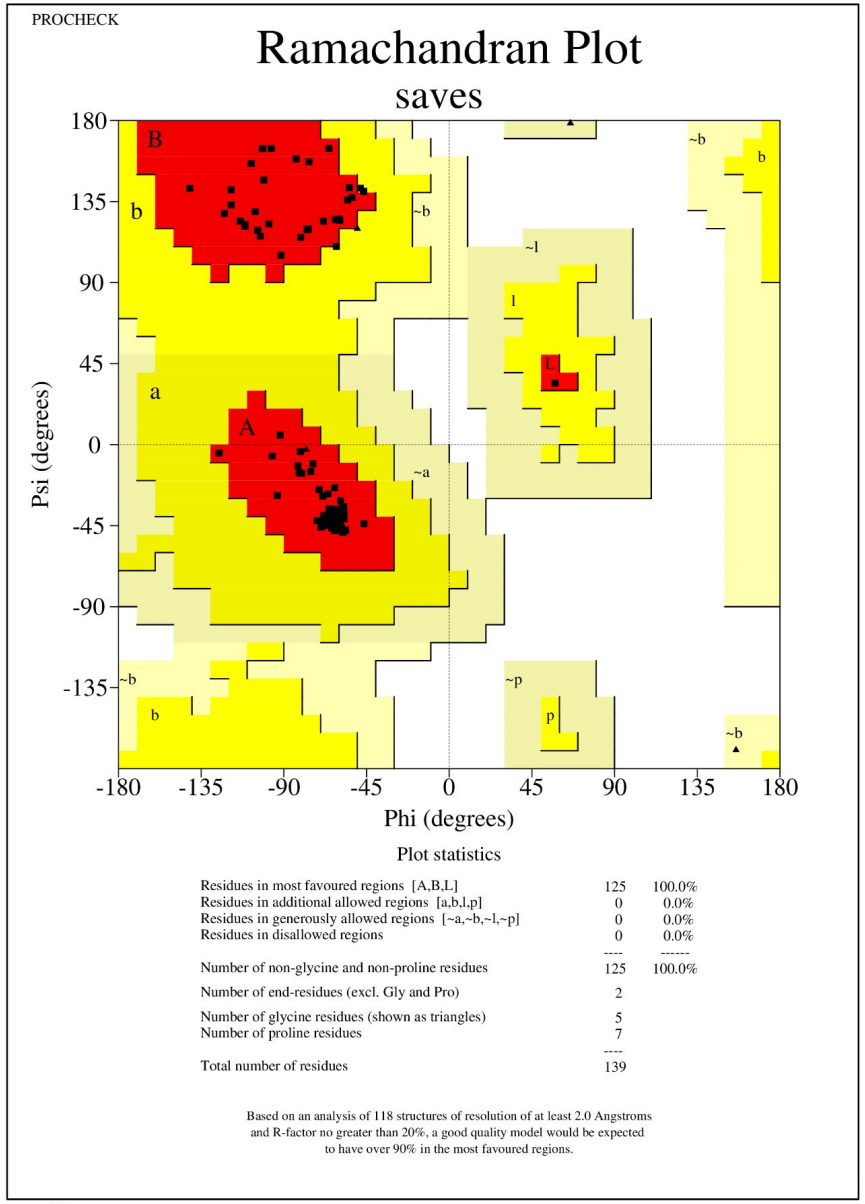
Model-Template Alignment



Quality Assessment

- Tools: PROCHECK, ERRAT, and Qualitative Model Energy Analysis (QMEAN).
- Assessments of the structure's accuracies and inaccuracies.

PROCHECK



saves_01.ps

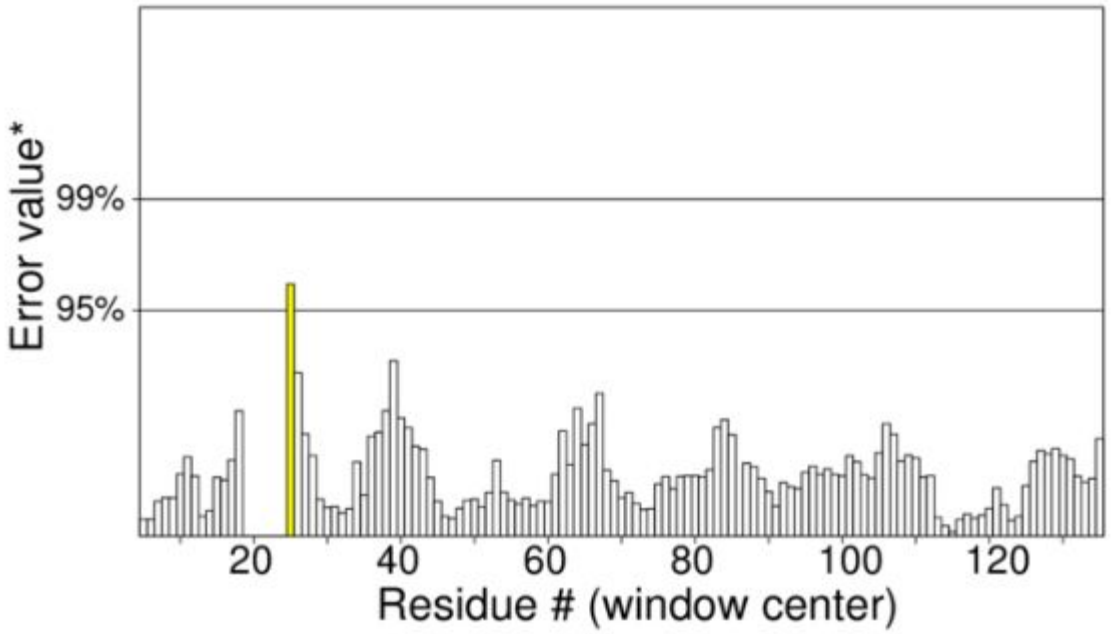
Plot statistics

Residues in most favoured regions [A,B,L]	125	100.0%
Residues in additional allowed regions [a,b,l,p]	0	0.0%
Residues in generously allowed regions [~a,~b,~l,~p]	0	0.0%
Residues in disallowed regions	0	0.0%
	---	------
Number of non-glycine and non-proline residues	125	100.0%
Number of end-residues (excl. Gly and Pro)	2	
Number of glycine residues (shown as triangles)	5	
Number of proline residues	7	

Total number of residues	139	

ERRAT

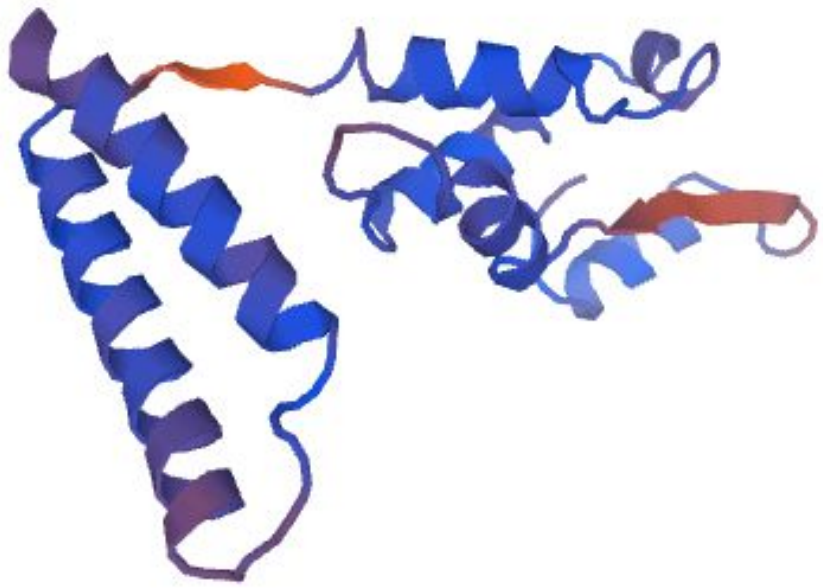
Program: ERRAT2
File: model_01.pdb
Chain#:A
Overall quality factor**: 99.194

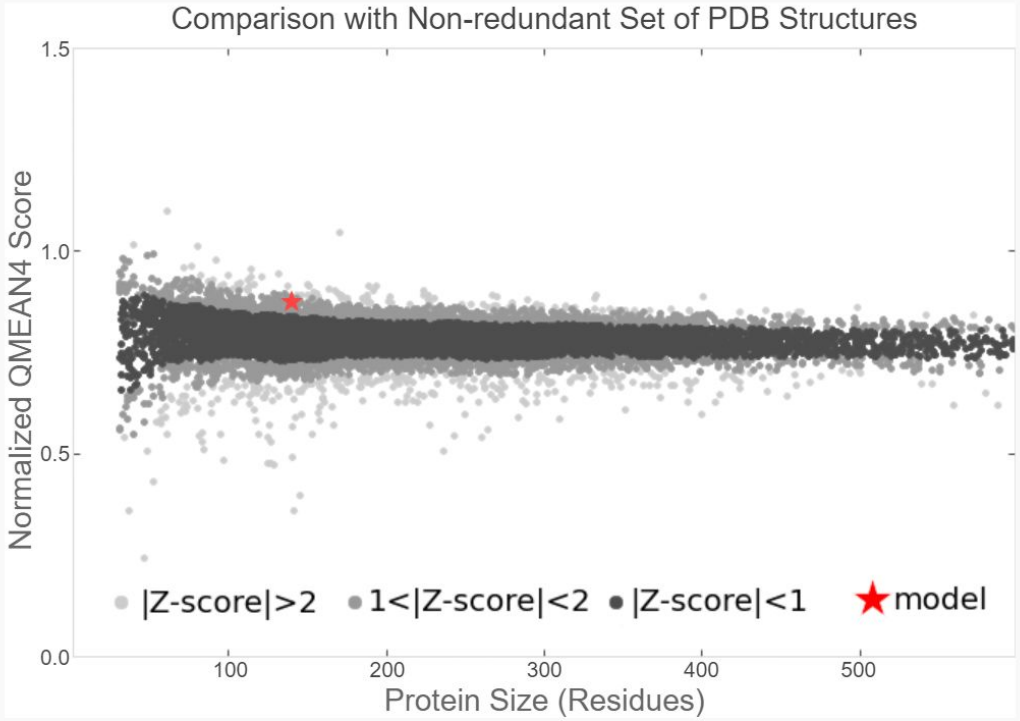


*On the error axis, two lines are drawn to indicate the confidence with which it is possible to reject regions that exceed that error value.

**Expressed as the percentage of the protein for which the calculated error value falls below the 95% rejection limit. Good high resolution structures generally produce values around 95% or higher. For lower resolutions (2.5 to 3A) the average overall quality factor is around 91%.

QMEAN

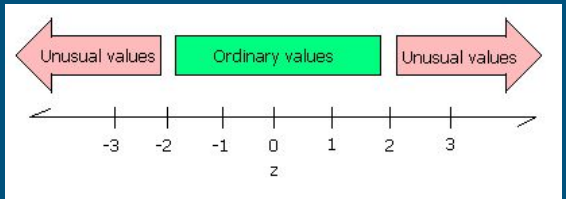




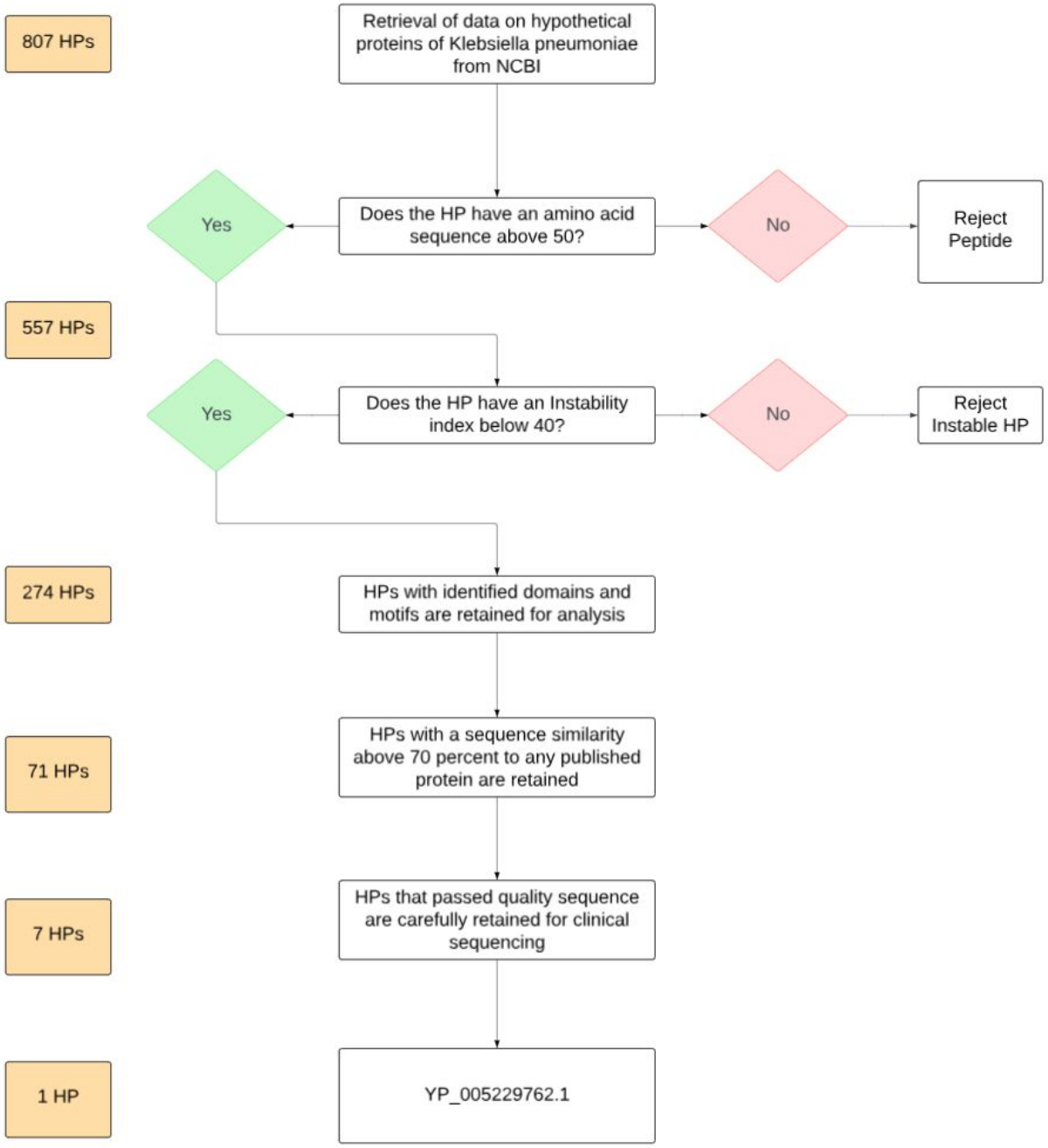
QMEAN4 Value: 1.83

Sequence colored by local quality:

```
A: MYGTREDLCRMLNEQYPAETPLNLIIWTPADIEALADGMEYSFSEHDVRAVLERMDTIPE  60
A: EQRLESGVSAGLVMALIDQVKENGQRVTVPVDLLETLLITAEQALWDREWTARDRNLPVP 120
A: ESVMRRLADTAKVRALLKS                                           139
```

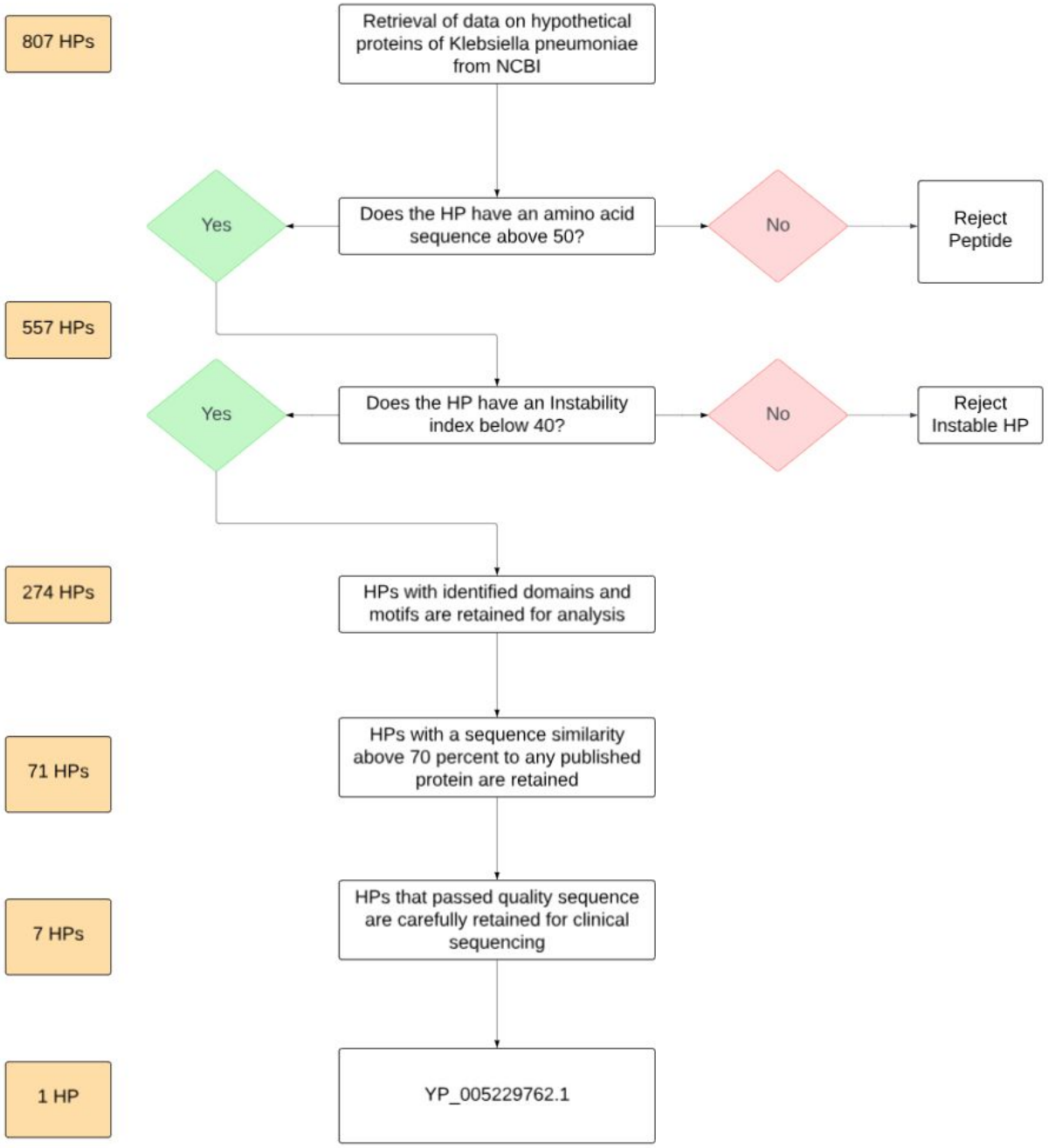


Flowchart



7 HPs

Flowchart



7 HPs → **1 HP**

Discussion/Conclusion

- Interpretation of results.
- Significance of findings in relation to DUF1380 Super Family.
- Potential implications for antibacterial agent development.

YP_005229762.1

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